Linear Algebra

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Source Code & Contributions
GitHub: https://github.com/sihyeon8240/math
Feel free to open issues or pull requests for corrections.

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Preface

Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \lor Q) \land R$$

  is evaluated by first computing $P \lor Q$.

- The **logical connectives** are interpreted in the order

$$\neg \; > \; \land \; > \; \oplus \; > \; \lor \; > \; \Longrightarrow, \; \Longleftarrow \; > \; \Longleftrightarrow$$

  Hence

$$P \oplus Q \implies \neg R \lor S \land T \iff U \impliedby V$$

  is understood as

$$((P \oplus Q) \implies ((\neg R) \lor (S \land T))) \iff (U \impliedby V)$$

- A **quantifier** binds the entire formula that follows it. Thus

$$\forall x, P(x) \land Q(x) \implies R(x)$$

  means

$$\forall x \, ((P(x) \land Q(x)) \implies R(x))$$

- The symbol $\Downarrow$ is **not** a logical connective; it is used only to indicate the flow of a proof or an explanation. For example,

$$\begin{array}{c} P \\ \Downarrow \\ Q \end{array}$$

  merely expresses that P, and therefore Q in the current discussion; it is not itself a proposition.

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Bibliography

[1] Serge Lang, *Linear Algebra*, Springer Nature.